

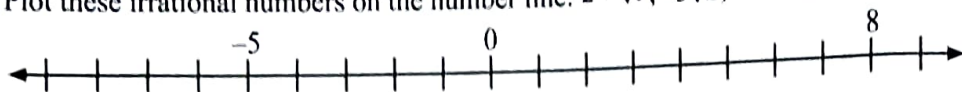
Chapter Check-Up

Multiple Choice Questions

- The square root of every positive non-perfect square number is a/an _____.
 a. positive number b. negative number c. irrational number d. zero
- Which of the following is an irrational number?
 a. $\sqrt{49}$ b. 1.555... c. $0.4\bar{5}$ d. $\sqrt{10}$
- What is the decimal representation of the irrational number π (pi) up to three decimal places?
 a. 2.718 b. 3.142 c. 1.414 d. 1.732

Practice Time

- Simplify.
 a. $2\sqrt{5} + 4\sqrt{5}$ b. $10\sqrt{3} - 3\sqrt{3}$ c. $\sqrt{11} \times \sqrt{7}$ d. $2\sqrt{3} \times 5\sqrt{2}$ e. $7\sqrt{21} + \sqrt{7}$
- Draw a number line and by geometrical construction mark $\sqrt{7}$ on the number line.
- Find the rationalising factors of the following irrational numbers:
 a. $-2\sqrt{8}$ b. $\sqrt{17}$ c. $3\sqrt{15}$ d. $4\sqrt{5}$ e. $7\sqrt{3}$
- Plot these irrational numbers on the number line: $2 + \sqrt{5}$, $-3\sqrt{2}$, $-8.1134...$, $\pi + 3$, $\sqrt{2} + 6$



- Find whether x and y are rational or irrational in the following.
 $a = 2 + \sqrt{5}$, $b = 2 - \sqrt{5}$; $x = a + b$, $y = a - b$
- Illustrate the closure property of addition of real numbers using the irrational numbers $\sqrt{7}$ and $2\sqrt{7}$.
- Is addition of real numbers commutative? Provide an example to illustrate your answer.

Case-Study Based Question

- Somya has a square piece of cardboard with an area of 36 sq. cm. She wants to find the length of each side of the square.
 Use the above information to choose the correct option.
 - What does Somya need to calculate to find the length of each side of the square cardboard?
 i. Perimeter ☒ ii. Area iii. Diagonal iv. Volume
 - Somya calculates that each side of the square cardboard is 6 cm long. What is the square root of the area of the cardboard?
 i. $\sqrt{6}$ ii. $\sqrt{12}$ ☒ iii. $\sqrt{36}$ iv. $\sqrt{72}$
 - Is the square root of the area of the given cardboard a rational or irrational number?
☒ i. Rational ii. Irrational
 iii. It depends on the size of the square iv. Cannot be determined

Everyday Maths

- Sumit needs to install a circular flowerbed with a radius of 7 m in his garden. Calculate the area of the circular flowerbed using the formula $A = \pi r^2$, where π (pi) is approximately 3.14159. Is the area a rational or irrational number? Provide the answer to three decimal places.

13. For a school project, Raghav builds a right triangle-shaped ramp with one leg measuring 3 m and the other leg measuring 4 m. Calculate the length of the hypotenuse using the Pythagorean theorem ($c^2 = a^2 + b^2$). Is the length of the hypotenuse a rational or irrational number?

Cross-Curricular Connect



C-10.1

(Sports) Javelin is a track and field event in which athletes throw a spear-like object, known as a javelin, for distance, aiming to achieve the longest throw possible. It requires a combination of technique, strength, and precision. Neeraj Chopra is an Indian athlete and Olympic gold medallist in javelin throw, making history by securing India's first track-and-field gold at the 2020 Tokyo Olympics. He is celebrated for his remarkable talent and dedication to the sport.



In a javelin-throwing competition, Priya wants to achieve a distance of exactly 50 m to secure a win. She knows that her personal record for the javelin throw is $2\sqrt{576}$ m. To reach her goal, she decides to work on her technique and practice.

- Calculate the additional distance, in metres, Priya needs to throw beyond her personal record to reach her goal of 50 m. Is this additional distance a rational or irrational number?
- In one of her practice throws, Priya achieves a distance of 49.8 m. How close is she to her goal? Calculate the difference in metres and express it as a decimal. Is this difference a rational or irrational number?

Being Indian



The Statue of Unity—a colossal monument dedicated to Sardar Vallabhbhai Patel, who was the first deputy prime minister and home minister of independent India—is in the Indian state of Gujarat, near the city of Vadodara. It stands in the Narmada district of Gujarat, along the banks of the Narmada River. Standing at 182 m, it is the world's tallest statue and a symbol of unity and national pride.

- The height of the Statue of Unity is 182 m. Is this height a rational or irrational number? Explain your answer.
- Calculate the square root of the height of the Statue of Unity and express it as a decimal to three decimal places. Is this value rational or irrational?

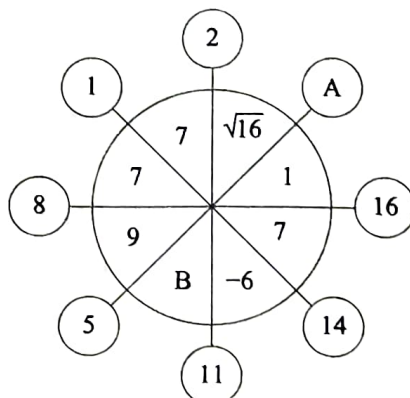


Brain Teaser



Observe the given figure. What integers should replace A and B?

C-7.2



Multiple Choice Questions

- Number of factors of $(a + b)^2$ is _____.
a. 1 b. 2 c. 3 d. 4
- The factorised form of $4p - 24$ is _____.
a. $4p \times 24$ b. $4(p - 8)$ c. $24(p - 4)$ d. $4(p - 6)$

Practice Time

- Find the perimeter of each of the following triangles whose sides are given by:

a. $2x + 3y + 3z$, $4x - 4y + 2z$, $6x + y - 2z$ b. $7x^2 + 6x + 3$, $8x^2 - 4x + 7$, $6x^2 - 4x + 9$

- How much is $4x^2 - 3x + 5$ smaller than $7 + 5x^2 - 3x$?

- Simplify the following.

a. $(-x^2) + (-x^2) + (-6x^2)$ b. $(4 + x) - (4 - x)$ c. $3(m - 6) + 2(4 + 5m)$ d. $-(-2x - 5y) - 4(2y - 4x)$

- Find the value of each of the following expressions when $p = -1$, $q = -2$, $r = 3$, $s = -2$.

a. $2p + 2r$ b. $(p - 3)(s + r)$ c. $(p + q)(p^2 + 3)$

- Find the product.

a. $4x(3x - 5)$ ~~b.~~ $8xy(x - y + 3)$ ~~c.~~ $10pq(1 - 12pq + p^2q^2)$ d. $\frac{1}{4}ab\left(\frac{1}{3}ab - 1\right)$
e. $a^2b^2\left(\frac{3ab}{4} + \frac{ab^2}{3} + 1\right)$ f. $(5x - 2y)(8x + 3y)$ g. $(z + 3b)(z + 6b)$ h. $(p^2 + pq + q^2)(p - q)$

- Simplify.

a. $\frac{a^{42}}{a^{20}}$ ~~b.~~ $\frac{-x^2}{-x^2}$ ~~c.~~ $\frac{63a^3b^2c^5}{-7a^2b^2c^2}$ d. $\frac{-54g^2h^4}{9gh}$ e. $\frac{120x^7y^8z^9}{40x^4y^3z^2}$

- Simplify.

a. $\frac{20a - 20b}{10}$ ~~b.~~ $\frac{-49y^2 - 14x^3}{-7x^2}$ ~~c.~~ $\frac{20m^{10} - 5m^3}{-5m^3}$ d. $\frac{-34m^5p^{12} - 17m^4p}{17m^2p}$

- Divide.

a. $(x^2 + 6x + 8) \div (x + 2)$ ~~b.~~ $(x^2 + 10x + 25) \div (x + 5)$ ~~c.~~ $(x^2 - 14x + 49) \div (x - 7)$
d. $(x^2 - 12x + 36) \div (x - 6)$ e. $(x^2 + 10x - 24) \div (x - 2)$ f. $(x^2 - 9x - 90) \div (x + 6)$

- Give the product of the following using the identity $(x + a)(x + b) = x^2 + (a + b)x + ab$.

a. $(x + 10)(x - 3)$ ~~b.~~ $(x - 3)(x - 5)$ ~~c.~~ $(x - 9)(x + 6)$ d. $(x - 7)(x + 5)$ ~~e.~~ $(x + 7)(x - 7)$
f. $(p + a)(p + b)$ g. $(x - 3)(x + 7)$ h. $(x - 8)(x - 2)$ i. $(p + 3)(p - 6)$

- Find the products using the identity $(a + b)^2 = a^2 + 2ab + b^2$.

a. $(2p + 4y)^2$ ~~b.~~ $(9x^2 + 4y^2)^2$ ~~c.~~ $\left(x + \frac{1}{x}\right)^2$ ~~d.~~ $(105)^2$
e. $(x + c)^2$ f. $(x + y)^2$ g. $(x + 3y)^2$ h. $\left(\frac{a}{b} + 1\right)^2$

13. Find the products using the identity $(a - b)^2 = a^2 - 2ab + b^2$.

- a. $(x - 4y)^2$ b. $(4x - 3y)^2$ c. $(2xy - 1)^2$ d. $\left(p - \frac{1}{p}\right)^2$ e. 98^2
 f. 390^2 g. $(x - b)^2$ h. $(2p - 3y)^2$ i. $\left(x - \frac{1}{x}\right)^2$ j. 59^2

14. Find the products using the identity $(a + b)(a - b) = a^2 - b^2$.

- a. $\left(\frac{x}{y} + \frac{y}{x}\right)\left(\frac{x}{y} - \frac{y}{x}\right)$ b. $\left(\frac{x}{2} + 1\right)\left(\frac{x}{2} - 1\right)$ c. $(x^4 + y^4)(x^4 - y^4)$ d. $\left(p + \frac{1}{p}\right)\left(p - \frac{1}{p}\right)$
 e. $(x + p)(x - p)$ f. $(3x - 4y)(3x + 4y)$ g. $\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)$ h. 105×95

15. $(32x^2 - 16x + 48)$ kilograms of sugar is stored in 8 bags in equal quantities. How many kilograms of sugar are there in each bag?

16. If $(x - 5)$ notebooks cost ₹ $(x^2 - 13x + 40)$, what is the cost of one notebook?

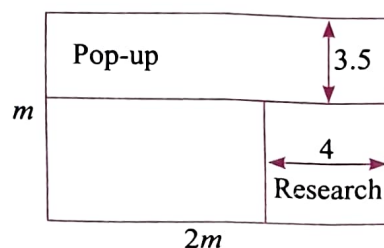
Case-Study Based Question

17. Let ₹ m be the pay of a worker on an hourly basis and for every hour of overtime, the pay is twice the normal pay. If the daily working hours are 8 hours and Mayank works 2 hours of overtime, then answer the following:

- a. Write a polynomial describing monthly wage of Mayank considering he works for 26 days.
 b. If the hourly pay is ₹100, find his total earnings for the month.

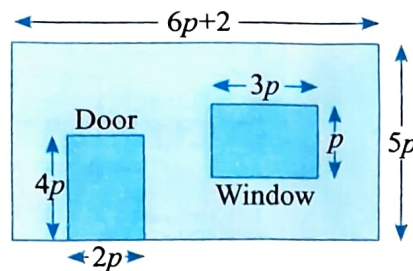
Creative Thinking

A link when clicked directs to a webpage which has a pop-up advertisement of height 3.5 inches. Towards the right is the research menu which is 4 inches wide. The figure is as shown alongside. Give an expression for the area of the webpage excluding the advertisement and the menu.



Everyday Maths

The figure shows the dimensions of a wall having a window and a door of a room. Write an algebraic expression for the area of the wall to be painted.



Mental Maths

1. If $x = 2$ and $y = -8$, find the value of $(x - y)^3$. 2. Find 103×97 .
 3. State the degrees of the four terms of this polynomial. Also, state the degree of the given polynomial.
 $7x^3y^3 - 45x^2y^2z + 8x - 15$

PA



Multiple Choice Questions

- What is the HCF of $20a^2b^3c^4$, $8a^2b^3c^5$, and $12a^3b^2c^3$?
 a. $12a^3b^2c^5$ b. $8a^3bc^2$ c. $12ab^2c^3$ d. $4a^2b^2c^3$
- The factorised form of $(15pq + 15 + 9q + 25p)$ is
 a. $(5p + 3)(3q + 5)$ b. $(5p - 3)(3q + 5)$ c. $(5p + 3)(3q - 5)$ d. $(5p - 3)(3q - 5)$

Practice Time

- Factorise the following.
 a. $50p + 50q$ b. $ax - bx$ c. $x - x^2$ d. $14a^3 - 22a$
 e. $7x - 14y$ f. $12y^5 - 12y^{12}$ g. $3cd - 3cf - 3ch$
- Find the factors.

		Middle term (sum of the numbers)	Product	Numbers	Factors
a.	$t^2 - 25t + 150$				
b.	$x^2 - 9xy - 36y^2$				
c.	$m^2 - m - 90$				
d.	$a^2 - 5a - 66$				
e.	$x^2 - 17x + 70$				

- Factorise the following.
 a. $4a^2 - 4a + 1$ b. $m^2 - 4m + 4$ c. $m^2 + 14m + 49$ d. $9x^2 - 30x + 25$
 e. $x^4 - y^4$ f. $x^4 - 16$ g. $k^2 - \frac{81}{121}$ h. $169a^2 - 52ab + 4b^2$
 i. $\frac{a^2}{4} - \frac{b^2}{25}$ j. $(a - 5)^2 - (b - 8)^2$ k. $27m^7 - 48m$ l. $81 - 9a^2 - 12ab - 4b^2$
- Factorise the following expressions.
 a. $x^2 - 6x - 72$ b. $x^2 - 25x + 150$ c. $x^2 - 5x - 300$ d. $x^2 + 15x + 54$
 e. $x^2 - 18x + 80$ f. $x^2 + 2x - 15$ g. $x^2 - 4x - 77$ h. $x^2 - 11x - 26$
 i. $x^2 - 16x + 39$ j. $x^2 + 20x + 36$ k. $x^2 - 10x - 24$ l. $x^2 - 2x - 8$
 m. $x^2 - 30x - 99$ n. $x^2 + x - 90$ o. $p^2 - 31p - 32$
- Factorise.
 a. $5x^2 - 32xy + 12y^2$ b. $2x^2 + 5xy - 12y^2$ c. $10a^2 - 83a - 17$ d. $4x^2 + 13xy + 10y^2$
- Factorise the following.
 a. $1 - 2x + x^2$ b. $24 + 11q + q^2$ c. $x^2 - 30x + 225$ d. $x^2 + 8x - 65$
 e. $x^2 + 19x - 66$ f. $16x^2 - 25$ g. $8x^2 - 10x - 3$ h. $8x^2 + 22x + 15$
 i. $(4p + 5q)^2 + 8(4p + 5q) + 16$ j. $(3x - y)^2 - 4(3x - y) + 4$

Case-Study Based Question

- Mahira got a scratch-and-win ticket on purchasing some groceries from a supermarket. She won some cash on the ticket and she also won ₹70 as a bonus. When she arrived at the cash counter, the person at the counter told her that she has won a 'triple your winnings' ticket because her bill amount is more

than ₹2500. As he was preparing the bill, he also told her that she was the 100th customer of the day, which automatically doubled her total winnings.

Based on the given information, answer the following questions:

- a. If Mahira won ₹ x through the scratch-and-win ticket, then the total amount won by her can be expressed as ₹ _____.
 - i. $2(x + 70)$
 - ii. $6(x + 70)$
 - iii. $6 + x + 70$
 - iv. None of these
- b. If Mahira's bill amount was less than ₹2500, then she would have won ₹ _____.
 - i. $2(x + 70)$
 - ii. $x + 70$
 - iii. $2x + 70$
 - iv. None of these

Everyday Maths

10. If Shalini spent an amount of ₹ $(ac^2 + 4ab^2 - 4abc)$ for buying $(c - 2b)$ same-priced shirts, find the price of each shirt.
11. To make a rectangular lawn in front of the school, the principal spent ₹ $(10x^2 + 35x + 30)$ at the rate of ₹5 per square metre. What is the area of the lawn? What are its dimensions?

Higher Order Thinking Skills

If the sum of $(x + 9)$ observations is $(x^4 - 6561)$, what is the mean of the observations.



Did You Know?

During a meeting of the American Mathematics Society in October 1903, F. N. Cole, the American mathematician, delivered a lecture entitled 'On the Factorisation of Large Numbers'.

Without a word, he approached the blackboard and wrote: $2^{67} - 1$, the 67th of the so-called Mersenne numbers, which Mersenne had claimed to be prime (divisible only by themselves and the number 1).

Next, Cole calculated (by hand) the value of 2^{67} and carefully subtracted 1 to yield: 147573952589676412927.

He then multiplied two numbers, again by hand: $193707721 \times 761838257287$. The result: 147573952589676412927.

Cole then silently returned to his seat. This is said to be the only talk ever given during an AMS meeting following which the audience applauded. There were no questions.

[Working every Sunday, Cole took about three years to find this factorisation.]

Brain Teaser



The length of a piece of rectangular floor tile is 5 cm more than twice the side x cm of a second square piece of tile. The width of the rectangular piece is 2 cm less than twice the side of the square piece. Find the area of the rectangular piece in terms of x (in expanded form).

C-72



Multiple Choice Questions

1. If a whole number is substituted for a , which expression is greatest?

a. $a + 9$

b. $a - 9$

c. $a - 10$

d. $a + 10$

Practice Time

2. Solve for the unknown.

a. $\frac{3}{-5}x = 15$

b. $\frac{4}{5}x = \frac{12}{25}$

c. $\frac{5}{8}x = 2.5$

d. $\frac{-13y}{2} = 2 \times 6$

3. Form an equation and solve.

a. $\frac{2}{3}$ of the total marks is 60. What are the total marks?

b. A man spends $\frac{2}{5}$ of his income, i.e., ₹3000 on his food. What is his income?

c. $\frac{2}{3}$ of the students of the school are boys. If the number of boys is 800, what is the number of students?

4. Solve the following.

a. $x + 7 = -17$

b. $5x + 3 = 23$

c. $8x - 7 = -23$

d. $16a - 3 = -5$

e. $\frac{x}{5} + 9 = 13$

f. $\frac{t}{3} - 5 = -6$

g. $\frac{2x}{3} - 15 = -7$

h. $5x + 8(2x - 9) = 54$

5. Solve.

a. $3x + 4 = 2x + 11$

b. $\frac{x-1}{2} - \frac{x-2}{3} = \frac{x-4}{7}$

c. $5(x - 13) = 10x - (2x + 5)$

d. $4p - 0.5 = 5p - 0.5$

e. $\frac{7}{2}(y - 1) = 4 + 3y$

f. $x - \left(2x - \frac{5x-1}{3}\right) = \frac{x-1}{3} + \frac{1}{2}$

6. Find three consecutive even numbers such that the sum of the first and the last numbers exceeds the second number by 10.

7. How many kilograms of tea, worth ₹300 per kg should be mixed with 10 kg of tea worth ₹240 per kg to produce a mixture to cost ₹260 per kg?

Case-Study Based Question

8. Ankita started a handicraft business with an investment of ₹7,14,600. The monthly operating cost of her business is given in the table.

Operations	Expenses
Rent	₹20,000
Electricity	₹5000
Overheads	₹40,000
Average salary per person	₹15,000

She hired six people to run the operations and was able to earn only operating cost in the first four months. From the fifth month onwards, she earned an average of ₹50,000 per month above the operating cost.

- After how many months, did she recover her investment?
- Ankita hired some more employees once she recovered her investment. Write an algebraic expression to show her operating cost after hiring.
- If her operating cost after hiring is ₹2,15,000 per month, find the total number of employees she has.

Everyday Maths

- An officer got an increase in salary by ₹2450 on his promotion. Now he has a salary of ₹27,000. What was his earlier salary?
- A newspaper boy got 42 new customers this month. He is now distributing to 673 customers. How many customers did he have before?
- In a purse, there are ₹20 notes, ₹10 notes, and ₹50 notes. The number of ₹50 notes exceeds two times the ₹10 notes by one. The number of ₹20 notes are 5 less than the number of ₹10 notes. If the total value of the money in the purse is ₹860, find the number of each variety of notes.

Higher Order Thinking Skills

- If a whole number is substituted for a , which expression is greater?
 - $a + 9$ or $a + 10$
 - $a - 9$ or $a - 10$
 - $a - 10$ or $a + 9$
- If m is the price marked on a pair of jeans, d is the discount on it, and s is the selling price (after discount) write an equation involving m , d , and s .
- $3a = 100$ and $3b = 200$. Which is greater $-a$ or b ?
- With what number should you multiply both sides of the equation to solve it?
 - $\frac{3}{4}s = 36$
 - $\frac{2}{3}n = 26$
- What is wrong with this solution?
 $6x - 24 = 30$
 $\Rightarrow \frac{6x}{6} - 24 = \frac{30}{6}$
 $\Rightarrow x - 24 = 5$
 $\Rightarrow x - 24 + 24 = 5 + 24$
 $\Rightarrow x = 29$

Brain Teaser



A machine operates in five stages as given below: **C-72**

Stage 1: Enter a number into the machine.

Stage 2: The machine doubles the number entered in Stage 1 and gives an output.

Stage 3: To the output of Stage 2, 9 is added and a new number is generated.

Stage 4: The output of Stage 3 is multiplied by 4 and a new number is generated.

Stage 5: The output of Stage 4 is printed on a ticket.

- Based on the information given above, complete the given table:

S.No.	Number entered at Stage 1	Number printed on the ticket at Stage 5
i.	4	
ii.	44	
iii.	-4	
iv.	0	

- Can you derive the expression on which the five stages of this machine are based?
- Say, we swap the positions of Stage 2 and Stage 3. Now, if we enter 7 at Stage 1, then what will be the difference between the numbers printed on the ticket at Stage 5 of both the old and new operations?

Chapter Check-Up



Multiple Choice Questions

- If marked price of an article is ₹3600 and the discount is 12% then the selling price of the article is _____.
 a. ₹3168 b. ₹3618 c. ₹3212 d. ₹3188
- If 90% of x is 630 km, then the value of x is _____.
 a. 650 km b. 700 km c. 700 m d. 650 m

Practice Time

- Find the profit percentage.

	CP	SP	Profit	Profit %
a.	₹6800		₹476	
b.		₹8064	₹864	
c.		₹8250	₹750	

- Find the loss percentage.

	CP	SP	Loss	Loss %
a.		₹468	₹52	
b.		₹437	₹23	
c.		₹6208	₹192	

- Find the unknown in each of the following.

- a. MP = ₹1780, Discount% = 15%, SP = ? b. SP = ₹1422, Discount% = 10%, MP = ?
 c. MP = ₹625, SP = ₹525, Discount% = ?

- The cost price of 10 pencils is the selling price of 8 pencils. What is the profit percentage?
- A man bought notebooks at the rate of ₹72 per dozen and sold them at the rate of ₹25 for a packet of 5 books. What was his gain or loss percentage?
- A man sells a chair for ₹3210 making a profit of 7%. What would have been his profit per cent if he had sold it for ₹3360?
- The marked price of a television is 25% more than the cost price. It is sold at a discount of 10%. If the marked price is ₹12,000, then find the cost price, selling price, and the profit. What will be the percentage of profit?
- Rashmi buys some winter clothes from the Winter Wear Shop at a discount of 20%. If the marked price for a pair of gloves is ₹186, a jacket is ₹839, and a sweater is ₹345, how much did she pay in all after discount?
- Afzal buys a television set whose marked price was ₹45,000. He then gets a discount of 15% and pays a sales tax of 12%. How much does he pay for the television set?

Case-Study Based Question

- A shopkeeper offers 10% discount on all his items. During the festival season, he decided to give a further discount of 20% over and above the existing discount.
 - Find the selling price of a set of frying pans of marked price ₹1000.
 - Find the marked price of a computer monitor that set was sold at ₹10,080.

Everyday Maths

- Amulya bought a second-hand car for ₹5,20,350. He polished the car for ₹2500, replaced old parts for ₹15,000, and then resold the car. He managed to make a profit of 14%. How much did he sell the car for?

14. A businessman buys 200 litres of vegetable oil at the rate of ₹60 per litre. He spends ₹2000 on packaging and ₹6000 on transport. He then sells 1 litre packets for ₹120. Does he make a profit? What is the profit or loss percentage?
15. A plot of land was bought for ₹4,00,000. The land value appreciates by 5% the first year and the value further appreciates by 8% the next year. What is the value of the land after two years?

Being Indian

Due to drought, the population of a village decreased by 12%, since families moved out of the village to find livelihood. If there were only 1584 people left in the village after the year, how many people inhabited the village before the drought?

Every year many parts of India face drought situations which force people to look for livelihood elsewhere. What do you think about this issue? Use Internet to find out the percentage of people in India that migrate due to drought every year?

Higher Order Thinking Skills

1. Arvind bought a dining table for ₹23,850. This includes 6% sales tax. The shopkeeper has made a profit of $12\frac{1}{2}\%$ on his investment. If the shopkeeper has spent ₹1000 on polishing and transportation, what is the actual cost of the table?
2. Gagan bought a plot of land for ₹2,35,000. He built a boundary wall around the plot and levelled the land and sold it for ₹35,00,000, thus making a profit of 40% on his investment. How much did he spend as overhead charges?
3. In a festive season, a retailer sells readymade cloths at a discount of 20% and still gains 20% of cost price. If the marked price for a pair of jeans is ₹4200, find the retailer's cost price.



Mental Maths

- | | | |
|------------------------------|-------------------------------|--------------------------------|
| 1. 10% of what number is 40? | 2. 4500 is 1% of what number? | 3. 250% of what number is 10? |
| 4. 1% of what number is 15? | 5. 30 is 20% of what number? | 6. 100% of what number is 423? |
| 7. 75% of what number is 66? | 8. 90 is 15% of what number? | 9. 50% of what number is 250? |

PA

Did You Know?

Every country has a central bank, which administers public finances and issues legal tender currency. One of the first such banks to be instituted was the **Bank of England**. It was founded in 1694 by 1268 shareholders to lend King William III money to finance his wars against France. In return it received 8% interest and the right to issue notes against the security of the loan. In India, the **Reserve Bank of India** issues legal tender notes.



Multiple Choice Questions

- The compound interest on ₹50,000 at 4% per annum for 2 years compounded annually is _____.
a. ₹4000 b. ₹4080 c. ₹4280 d. ₹4050
- A sum is taken for two years at 16% p.a. If interest is compounded after every three months, the number of times for which interest is charged in 1 year is _____.
a. 8 b. 6 c. 4 d. 9

Practice Time

- Find the simple interest and the amount.

S.No.	Principal	Rate % p.a.	Time	SI	Amount
a.	₹560	4%	73 days		
b.	₹2700	9%	146 days		
c.	₹10,000	12%	3.25 years		

- Find the principal.

S.No.	SI	Time	Rate p.a.	Principal
a.	₹392	3.5 years	3.5 %	
b.	₹4.50	5 months	4.5 %	

- Find the simple interest and the rate per cent.

Principal	Amount	SI	Time	Rate %
₹10,000	₹11,800		2 years	

- Find the amount and compound interest.

S.No.	Principal	Rate % p.a.	Time (years)	Amount	CI
a.	₹72,000	6%	3		
b.	₹5000	9%	2		

- Find the CI on:

- ₹7000 at 10% p.a. compounded annually for 4 years.
- ₹30,000 at 12% p.a. compounded half-yearly for 2 years.

- A sum of money amounts to ₹780 in two years and ₹1230 in seven years. Find the principal and rate of simple interest.

- In how much time will a sum of money double itself if invested at 8% simple interest per annum?

(Hint: Let the principal be ₹100. Interest will also be ₹100.)

- Rajan borrowed ₹5000 at 8% p.a. for 3 years. Find the difference between compound interest and simple interest, assuming compounded annually.

- Aashna deposited ₹12,000 in a bank for 2 years. It is compounded annually at 9% p.a. What amount will she receive on maturity (at the end of 2 years)?

- Find the principal if it amounts to ₹16,335 compounded annually at the rate of 10% p.a. in two years.

- In how many years will ₹8800 amounts to ₹10,648 at 10% p.a. compounded annually?

- At what rate % of CI will ₹4000 amount to ₹5324 in three years?

15. Calculate the compound interest on ₹20,000 for $1\frac{1}{2}$ years at the rate of 10% per annum compounded semi-annually.
16. Find the compound interest on ₹1,60,000 for 2 years at 10% per annum, compounded semi-annually.
17. The population of a town increases 4% annually. If the present population is 54,080, what was the population two years ago?
18. In the year 2019, the number of malaria patients admitted in the hospitals of a state was 4375. Every year this number decreases by 8%. Find the number of patients in 2021.
19. A sum of ₹15625 invested at 8% p.a. compounded semi-annually amounts to ₹18279.04. Find the time period.

Case-Study Based Question

20. Meera bought a used sewing machine for ₹8664. The first owner of the sewing machine handed over the original bill along with the item. According to the original bill it was purchased for ₹9600.
 - a. If the sewing machine is 2 years old, then its value depreciated at
 - i. 7% p.a.
 - ii. 4.5% p.a.
 - iii. 5% p.a.
 - iv. 6.2% p.a.
 - b. Meera sold the sewing machine after one year. If the rate of depreciation remains the same, then Meera lost _____ on the transaction.
 - i. ₹433.20
 - ii. ₹866.40
 - iii. ₹768.50
 - iv. ₹1020.55

Everyday Maths

21. The population of a town is 50,000. If the annual emigration rate is 6% and the annual immigration rate is 4%, calculate the population after two years.
22. The present cost of a mobile phone is ₹15000. If its value decreases every year by 5%, find its cost 2 years ago.

Being Indian

Blood donation is a noble cause. Every year the country sees around lakhs of blood donors. At a blood donation camp in an area 1500 people donated blood last year.

- a. If the number increased by 5% this year, how many people donated blood at the camp this year?
- b. If the number of donors is expected to rise by 8% next year, how many donors will the camp record next year?
- c. Is it important to donate blood? Why?

Higher Order Thinking Skills

1. The value of a car depreciates at the rate of 10% per year. A car which was bought three years back is now worth ₹4,73,850. What was its original price?
2. In fixed deposit, a bank gives 10% interest compounded annually, for senior citizens. How many complete years will it take for a sum of money to be more than double?
3. The difference between the simple interest and compound interest at the rate of 10% for two years is ₹1. What is the principal?



LUCA PACIOLI

Fra Luca Bartolomeo de Pacioli (1447-1517), an Italian mathematician, is regarded as the 'Father of accounting'. He described one of the first book keeping systems used by the Venetian merchants during the Italian Renaissance, now known as the double-entry accounting system. His system resembles most part of the accounting that we use today. He also touched upon the concepts of balance sheets and income statements. A dedicated text that he wrote for his students contained sections on merchant arithmetic, such as barter, exchange, profit, mixing metals, and algebra.

Multiple Choice Questions

- Which of the following vary inversely with each other?
 - speed and distance covered
 - distance covered and taxi fare
 - distance travelled and time taken
 - speed and time taken
- If x and y are inversely proportional then _____ = k , where k is positive constant.
 - xy
 - $x \div y$
 - $x + y$
 - $x - y$

Practice Time

- Complete the equivalent ratios.

a. $\frac{1}{3} = \frac{2}{\square} = \frac{3}{\square} = \frac{\square}{12}$

b. $\frac{3}{5} = \frac{6}{\square} = \frac{9}{\square} = \frac{\square}{20}$

c. $\frac{1}{8} = \frac{7}{\square}$

d. $\frac{13}{40} = \frac{39}{\square}$

e. $\frac{5}{6} = \frac{\square}{24}$

f. $\frac{6}{7} = \frac{\square}{14} = \frac{18}{\square} = \frac{\square}{28}$

- x varies directly as y^2 , $x = 7$ when $y = 1$, find x when $y = 3$.
 - y varies directly as x^2 , $x = 2$ when $y = 3$, find y when $x = 6$.
 - x varies directly as y^3 , $x = 2$ when $y = 1$, find x when $y = 3$.
- The number of girls and boys in a school is 525 and 675, respectively. What is the ratio of the boys to girls in its simplest form?
- The English textbook of Class VIII costs ₹240 and the Mathematics textbook costs ₹210. What is the ratio of the cost of the English textbook to Mathematics textbook in its simplest form?
- A recipe needs one cup of flour and $1\frac{1}{2}$ cups of sugar. If 3 cups of flour are used, how much sugar is needed?
- If five eggs cost ₹28, how many eggs can be bought for ₹224?
- The number of stamps collected by Arvind and Arshu are in the ratio 7:3. Together, they have 180 stamps. Find the number of stamps each one has.
- The pocket money of Joan and Joyce are in the ratio 10:8. If Joan has ₹20 more than Joyce, find the pocket money of each one.
- Two supplementary angles (two angles whose sum is 180°) are in the ratio 5:1. Find the measure of each angle.
- If we divide ₹5250 in the ratio 6:7:8 among three friends—Sarosh, Santosh, and Rajiv—how much will each of them get?
- Amarjeet earns ₹160 for three hours of work. How much will he earn for 27 hours of work?
- If $\frac{1}{4}$ kg of a sweet costs ₹52, find the cost of $2\frac{1}{2}$ kg of the same sweet.
- The cost of bus tickets for 5 people for a journey was ₹95. What is the number of tickets that can be bought for ₹342 for the same journey?
- If 16 men can make 320 art pieces in one day, how many men will be needed to make 960 such art pieces in one day?
- Sony and Ranu together can do a piece of work in 20 days. Sony alone can finish the work in 30 days. If Ranu would have been doing it alone, how long will she take?
- On a library shelf, 150 books, each of thickness 3 cm, can be stacked. If books, each of thickness 2 cm, are to be stacked in the same shelf, how many such books can be stacked?

Case-Study Based Question

19. A 400 m long train starts from city A at 5:00 a.m. and reaches city B at 6:00 a.m., covering a distance of 150 km.
- How long will it take to cross a bridge 800 m long bridge?
 - On a certain stretch moving at a speed of 120 km/h, the train passes a bridge in 21s. What is the length of the bridge?
 - How much time does another train, moving at a speed of 72 km/h and of half the length of this train take to pass this train standing at a platform?

Everyday Maths

20. A school buys 40 blackboards at ₹650 each. If each blackboard were to cost ₹150 less, how many blackboards in total could have been bought by the school with the same money?
21. Working for 8 hours daily, 40 people can dig the foundation of a building in 21 days. Working for 10 hours daily, if the work is to be finished in 14 days, how many people are needed to do the same work?
22. Roshan alone can build a fence in 20 days. He starts the work and leaves it after 5 days. Ramesh does the remaining work in 12 days. How long will it take if Ramesh did the whole work alone? How long will it take if Roshan and Ramesh work together from the beginning?

Higher Order Thinking Skills

A contractor calculated that he could finish the construction of a road in 40 working days. He knew that if he appointed 32 people and made them work for 6 hours a day for 5 days in a week, the work would finish in time. But he could arrange for only 20 workers. He decided to make them work for 8 hours a day for 6 days a week. Will they complete the work in time? If not, how many more days will they take?



Journal

How can you write 3:5 as decimal and percentage? If you were to describe the conversions, would you be right in using the word 'proportional'? Explain.

Did You Know?

Nowadays borrowing and lending money at an interest is a normal and accepted procedure. The whole concept of banking is based on the ability of capital to earn interest. However, this was not always so. In Europe in the Middle Ages, for example, lending money at an interest was considered a sin known as *usury*.

Christians were forbidden to lend money because the concept of earning money without having to work for it was regarded as immoral. Only the Jews were permitted to lend money and for this they were often unjustly associated with usury, extortion and greed. The play *The Merchant of Venice* is a play by William Shakespeare that revolves around the theme of usury.

Brain Teaser



C-72

Mona has a 2.5 in. $\times$ 4.25 in. picture which she wants to print in different enlarged sizes using an automatic digital printer. The manual of the printer lists the following standard sizes:

Mona knows that the printer will enlarge the picture to fit the size she chooses. So, if the standard size is not exactly proportional to the size of the picture, two things may happen: either the print will be distorted, or some portions of the picture will not be printed. Help her choose the sizes so that she gets perfectly enlarged, distortion-free photographs.

Regular sizes	Medium sizes	Large sizes
5.0 $\times$ 8.0	20.0 $\times$ 34.0	50.5 $\times$ 85.5
5.0 $\times$ 8.50	25.0 $\times$ 40.0	55.0 $\times$ 93.5
7.5 $\times$ 12.75	30.0 $\times$ 50.0	62.5 $\times$ 105.0

Chapter Check-Up

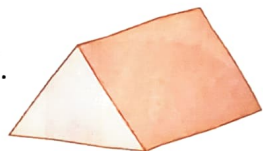


Multiple Choice Questions

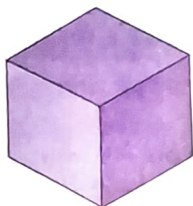
- Which of the following is a regular polyhedron?
a. Cuboid b. Triangular prism c. Cube d. Square prism
- In a solid, if $F = V = 5$, then the number of edges in this shape is _____.
a. 8 b. 4 c. 6 d. 2

Practice Time

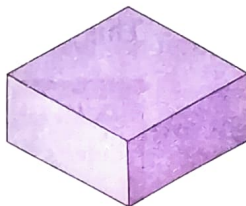
- Draw a 2-D representation of the solid shown here, showing hidden edges as well.
- Draw the top, front, and side view of the following figures. Also draw them on an isometric dotted paper and grid paper.



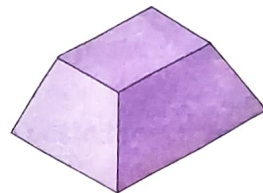
a.



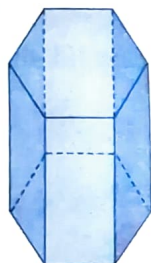
b.



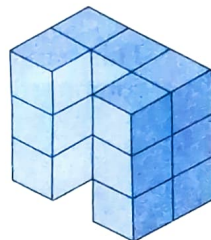
c.



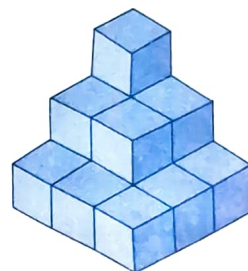
d.



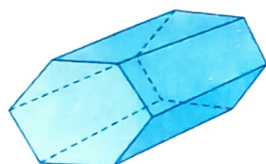
e.



f.

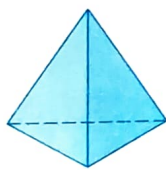


- Look at the solids drawn below and fill the chart given here. Check whether Euler's formula holds good in each case.



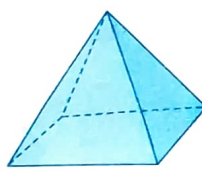
Hexagonal Prism

a.



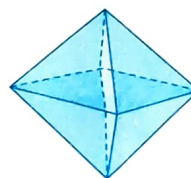
Triangular Pyramid

b.



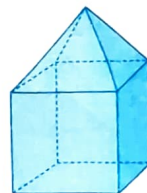
Square Pyramid

c.



Octahedron

d.



Nested Solid

e.

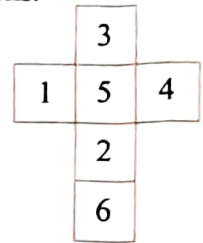
S.No.	Polyhedron	Faces (F)	Vertices (v)	Edges (E)	$(F + V) - E$
a.	Hexagonal prism				
b.	Triangular pyramid				
c.	Square pyramid				
d.	Octahedron				
e.	Nested solid				

- Using Euler's formula, find the unknown.

Faces	6	5	20	14
Vertices	x	y	36	w
Edges	12	9	z	36

Case-Study Based Question

7. Kiran made a net that can be folded into the shape of a cube. She marked the faces with numbers 1 to 6. The given figure shows the net made by Kiran. Observe the net and answer the following questions.



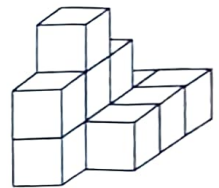
- The face marked with 1 is opposite to the face marked with ____.
i. 2 ii. 3 iii. 4 iv. 5
- The face marked with 5 is opposite to the face marked with ____.
i. 6 ii. 4 iii. 2 iv. 1
- If she was making a die, then did she mark the numbers correctly? Give reason. No.

Creative Thinking

Draw a map of any one floor of your school. Mark all necessary places like library, medical room, sports room, music room, dance room, classes, reception, etc.

Everyday Maths

Vikas is playing with interlocking cubes and has arranged them as shown alongside. Observe the arrangement and answer the following questions.



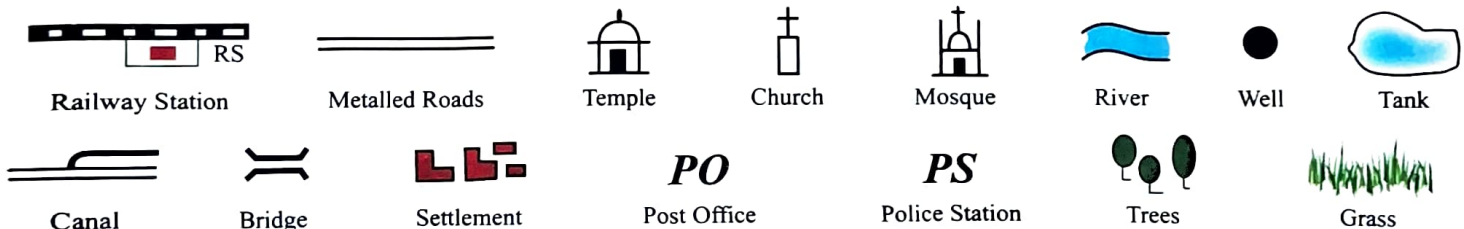
- How many cubes has he used?
- How many more cubes does he need to make a cubical arrangement?
- Draw the top, front, and side view of the given shape.

Cross Curricular Connect



C-10.1

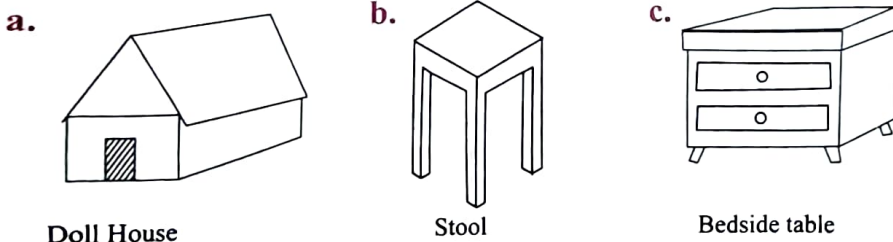
(Geography) Symbols are an important component of any map. It is not possible to draw the actual shape and size of different features such as buildings, roads, bridges, trees, railway lines, etc on a map. These symbols give a lot of information in a limited space. There is an international agreement regarding the use of these symbols. These are called **conventional symbols**. Some of the conventional symbols are shown below.



Create a map of your locality using some of the conventional symbols given above.

Being Indian

Every year during summer vacation, Tarun joins a workshop organised by his school. This year he has joined a furniture workshop and learned to make various furniture items. The line drawing of some of his works are given below. Draw the top, front, and side view of these items.



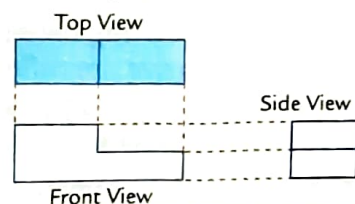
What value does Tarun show by joining the workshops every year?

Brain Teaser



C-7.2

Can you imagine how the given solid would look in three dimensions, just by looking at its top view, front view, and side view on paper?



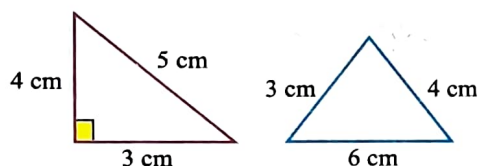
Multiple Choice Questions

- Two squares not necessarily of the same size are _____ figures.
 - similar
 - congruent
 - symmetric
 - equal
- In $\triangle PQR$ and $\triangle LMN$, $\frac{PQ}{MN} = \frac{PR}{ML}$ and $\angle P = \angle M$, then symbolically we write _____.
 - $\triangle PQR \sim \triangle LMN$
 - $\triangle PQR \sim \triangle MLN$
 - $\triangle PQR \sim \triangle MNL$
 - $\triangle PQR \cong \triangle MNL$
- In $\triangle PQR$ and $\triangle LMN$, $\frac{PQ}{LM} = \frac{QR}{NL}$, then they will be similar when _____.
 - $\angle Q = \angle M$
 - $\angle P = \angle L$
 - $\angle P = \angle N$
 - $\angle Q = \angle L$

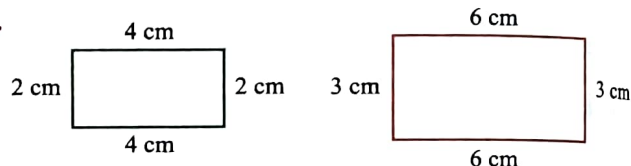
Practice Time

- State whether the following pairs of figures are similar or not?

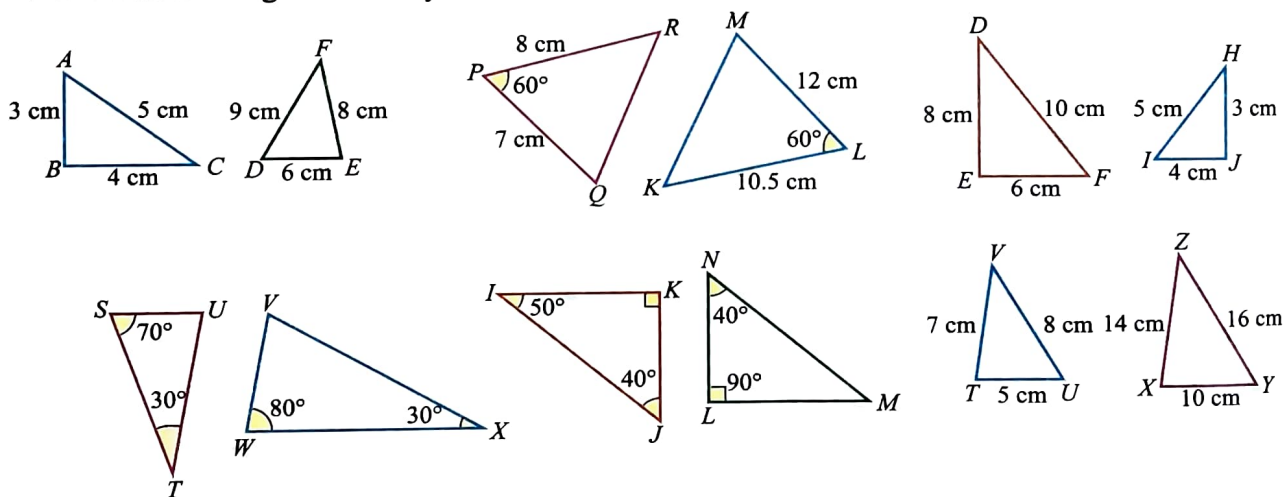
a.



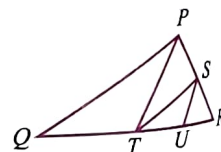
b.



- Which of the following triangles are similar? Also, state the similarity criterion used and write the pairs of similar triangles in the symbolic form.

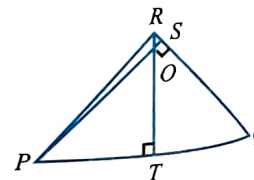


- In the given figure, $SU \parallel PT$ and $ST \parallel PQ$. Prove that $\frac{PR}{UT} = \frac{RT}{TQ}$.



- Using BPT, prove that a line drawn through the mid-point of one side of a triangle parallel to another side bisects the third side.

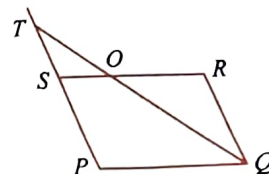
- Using the converse of BPT, prove that the line joining the mid-points of any two sides of a triangle is parallel to the third side.



- In the given figure, altitudes PS and RT of $\triangle PQR$ intersect each other at the point O . Show that:

- $\triangle PTO \sim \triangle RSO$
- $\triangle PQS \sim \triangle RQT$
- $\triangle PTO \sim \triangle PSQ$
- $\triangle OSR \sim \triangle QTR$

10. In the figure given alongside, T is a point on the side PS produced of a parallelogram $PQRS$ and QT intersects SR at O . Show that $\triangle PQT \sim \triangle ROQ$.
11. S is a point on the side QR of a triangle PQR such that $\angle PSR = \angle QPR$. Show that $RP^2 = RQ \cdot RS$.



Case-Study Based Question

12. Observe the following figures carefully and answer the following questions.

- a. Which of the given figures are congruent figures?
 i. Fig. a ii. Fig. b iii. Fig. c iv. Fig. d
- b. Which of the given figures are similar figures?
 i. Fig. a ii. Fig. b iii. Fig. c iv. Fig. d
- c. Which of the following statements is correct?
 i. All congruent figures are similar.
 ii. All similar figures are congruent.
 iii. The criterion for similarity and congruency is the same.
 iv. Similar figures have the same size and shape.



Fig. a

Fig. b

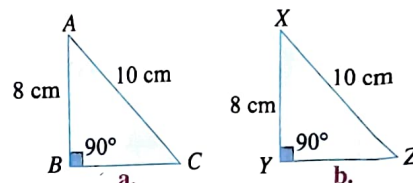


Fig. c

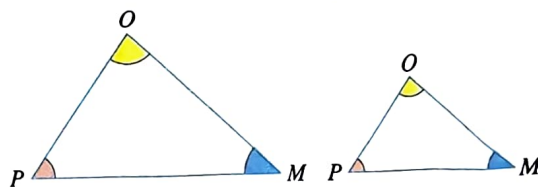


Fig. d

Everyday Maths

13. The concept of similar triangles is used to find long distances and heights of tall objects. There is a very tall building whose height needs to be found out. A pole of 5 m height was erected on some distance such that the pole was parallel to the height of the building. At 11 a.m., the shadows of the pole and the building were measured. It was found that the shadow of the pole was 3 m and that of the building was 30 m. What will be the height of the building using the concept of similar triangles here? Which similarity criterion will you use?

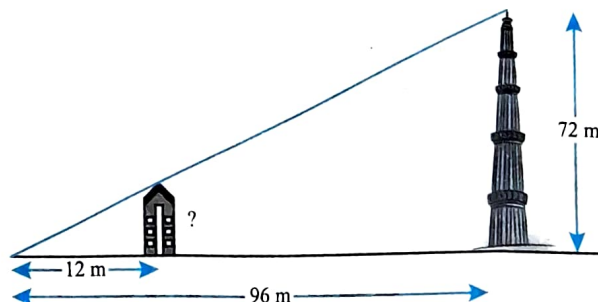
Cross-Curricular Connect



C-10.1

(Social Sciences) The Qutub Minar is a minaret in Delhi, India. It was built in the 12th century by Qutub-ud-din Aibak. The minaret is made of red sandstone and marble, and it is about 72 m tall. It is one of the tallest minarets in the world, and it is a UNESCO World Heritage Site. The minaret is covered in intricate carvings and inscriptions, and it is a beautiful example of the craftsmanship of the time. The Qutub Minar is a popular tourist destination, and it is one of the most iconic landmarks in Delhi.

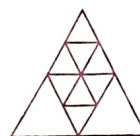
One afternoon the shadow of the monument is 96 m long and at the same time a building standing in the same straight line casts a shadow of 12 m, find the height of the building.



Brain Teaser



The given picture shows equiangular triangles. Can you find the total number of equiangular triangles in this picture?



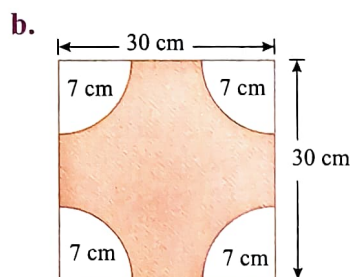
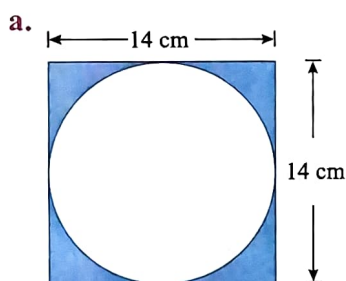
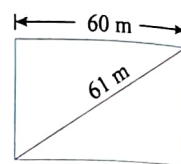
C-72

Multiple Choice Questions

- A regular hexagon is inscribed in a circle of radius a . The perimeter of the regular hexagon is
a. $3a$ b. $12a$ c. $9a$ d. $6a$
- The area of an equilateral triangle whose sides are 20 cm each is
a. 60 sq. cm b. $100\sqrt{3}$ sq. cm c. $20\sqrt{3}$ sq. cm d. 100 sq. cm

Practice Time

- The length and breadth of a rectangular field are in the ratio 5:3. The area of the field is 2160 sq. m. If a fence is to be made at the rate of ₹350 per metre, how much will it cost?
- The diagonal of a rectangular courtyard is 61 m. If the length is 60 m, find the area of the courtyard.
- Find the area of squares (in cm^2) whose diagonals are of the following lengths:
a. 75 cm b. 0.236 m c. 6.34 dm
- The area of a triangular billboard is 3.125 sq. m. If its base is 2.5 m long, what is its height?
- A right-angled triangle has a hypotenuse 13 cm long. One of its sides is 12 cm. What is its area?
- A triangular piece of land has one of its angles equal to 90° . The area of this land is 330 sq. m. If one side adjacent to the right angle is 11 m, what are the lengths of the other two sides?
- The length of one side of a parallelogram is 125 cm. The distance of this side to the opposite side is 60 cm. Find its area.
- The area of a parallelogram is 660 sq. m. One of its sides is 30 m long. What is the distance of this side to the opposite side?
- The area of an equilateral triangle is $64\sqrt{3}$ sq. m. Find the length of each side of the triangle.
- Find the area of the shaded portions in the following figures.

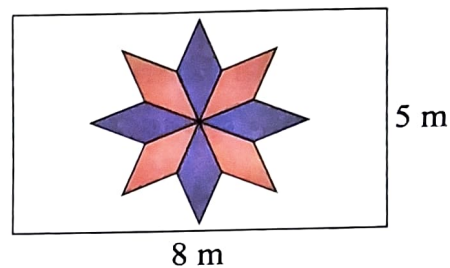


- Find the area of a trapezium, the lengths of whose parallel sides are 80 cm and 60 cm. The distance between these sides is 25 cm.
- The area of a trapezium is 897 sq. m. If the parallel sides are of length 32 m and 46 m, what is the distance between the parallel sides?
- The area of a trapezium is 1100 sq. m. The distance between two parallel sides is 20 m and one of the parallel sides is 70 m. Find the length of the other parallel side.
- Find the area of a quadrilateral, if the length of one of its diagonals is 56 cm and the perpendicular distance from the other two vertices to this diagonal are 30 cm and 42 cm.
- The diagonal of a square field is 90 m. What is its area?
- The area of a square garden is 450 sq. m. What is its length from one corner to the opposite corner?

Case-Study Based Question

19. The living room floor of a house is covered with white marble and has a flower design made from coloured stone. The design has 8 rhombus petals of same size, the diagonals of each petal are of length 1 m and 80 cm.

- The area of each petal is _____.
i. 4 m^2 ii. 0.4 m^2 iii. 40 cm^2 iv. 400 cm^2
- How much colour stone is needed for the design?
i. 32 m^2 ii. 320 m^2 iii. 3.2 m^2 iv. 0.32 m^2
- How much area is covered with white marble?
i. 40 m^2 ii. 36.8 m^2 iii. 32.6 m^2 iv. 44 m^2



Creative Thinking

A rectangular garden 200 m long and 150 m wide has a path all around it, on the inner side, having a width of 3 m. In the centre of this plot, there is a circular pond of radius 7 m. What area of the land is left for the lawn and the flower beds?

Everyday Maths

The length, breadth, and height of a hall are 10 m, 8 m, and 6 m, respectively. The hall has two doors of area $2 \text{ m} \times 1.5 \text{ m}$ and three windows of area $1.5 \text{ m} \times 1 \text{ m}$. Find the cost of painting the walls (only the walls) at the rate of ₹180 per square metre.

Cross-Curricular Connect



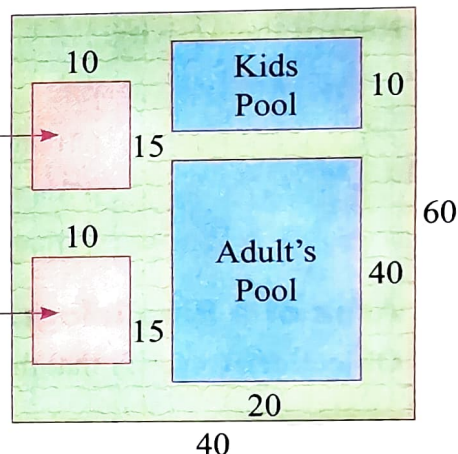
(Sports) Swimming is a healthy activity that can be done by most people throughout their life. It is a low-impact workout that has several mental and physical benefits, all while being a good recreational activity. Swimming builds endurance, muscle strength, and cardiovascular fitness. It is also known as the best full-body workout. Swimmer **Sachin Nag** became the first Indian to win a gold medal at the Asian Games by winning the 100 m freestyle gold at the inaugural event in New Delhi on March 7, 1951.

The layout of the swimming pool compound in a society is given alongside (all dimensions in feet).

- Find the area covered with green tiles.
- How many square tiles of side 2 feet are required to cover the area in part a?

Men's changing area

Ladies' changing area



C-10.1

Mental Maths

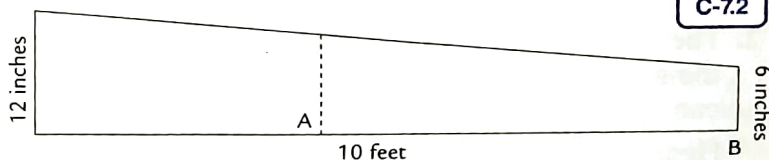
- If the measurements of sides of a square are doubled, what happens to the area?
- If the sides of a rectangle are doubled, what happens to the area?
- If the base and altitude of a parallelogram are doubled, what happens to the area?
- If the base and altitude of a triangle are doubled, what happens to the area?

PA

Brain Teaser



Raghav had a board measuring 10 feet in length, 6 inches wide at one end, and 12 inches wide at the other, as shown in the given figure. How far from B must the straight cut at A be made in order to divide it into two pieces of equal area?



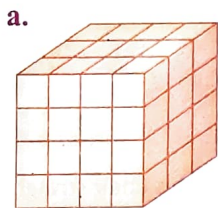
C-7.2

Multiple Choice Questions

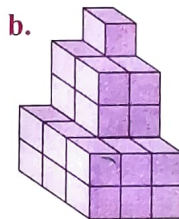
- The ratio of the lateral surface area of a cube to its total surface area is _____.
a. 3:2 b. 2:3 c. 1:2 d. 3:1
- What is the area of the largest triangle that can be fitted into a rectangle of length l units and width w units?
a. $\frac{lw}{2}$ b. $\frac{lw}{3}$ c. $\frac{lw}{6}$ d. $\frac{lw}{4}$

Practice Time

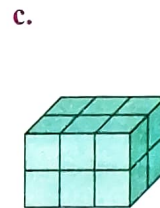
- A metallic cuboid of dimensions 60 cm, 40 cm, and 20 cm is melted and made into cubes of 2 cm side. How many such cubes can be obtained?
- The dimensions of a cuboid are in the ratio 5:2:1. Its volume is 1250 cubic metres. What is its total surface area?
- A log of wood of dimension 2 m $\times$ 20 cm $\times$ 10 cm is cut into small blocks of 10 cm $\times$ 5 cm $\times$ 4 cm. How many blocks will we get? What will be the total surface area of all these blocks?
- How much cardboard is needed to make a rectangular box of length 12 cm, breadth 10 cm, and height 7 cm?
- A wall is 8 m long, 4 m high, and 22.5 cm thick. How many bricks of length 25 cm, breadth 12.5 cm, and 8 cm high are needed to build the wall?
- The volume of a cube is 125 cm^3 . What is the measure of its side?
- If the total surface area of a cubical tank is 486 sq. m, what is the length of one side?
- If the length, breadth, and height of a cuboid are in the ratio 5:4:3 and its total surface area is 376 sq. m, find the dimensions of the cuboid.
- A total of 500 boxes of dimensions 30 cm by 20 cm by 20 cm are required for packing some books. How much of cardboard will have to be used for the same?
- Find the area of a tin sheet required to make a cylindrical vessel with lid if its radius of the base is 14 cm and the height is 16 cm. Also, find the capacity of the vessel.
- Find the volume of the following figures.



Volume _____



Volume _____



Volume _____

Case-Study Based Question

- A ghee manufacturing company decides to discontinue a cylindrical ghee container of diameter 14 cm and height 20 cm. The company plans to replace the container with tetra packs.
 - If the ghee of one cylindrical container is divided into four tetra packs of base area 40 cm^2 , the height of the tetra pack is _____.
i. 17.5 cm ii. 19.25 cm iii. 19 cm iv. 21.7 cm
 - They also decided to make larger tetra packs with base area 70 cm^2 and height 22 cm. How many of these tetra packs is equal one cylindrical container?
i. 2 ii. 3 iii. 2.5 iv. 3.5

Chapter Check-Up



Multiple Choice Questions

- The range of the data— 9, 8, 4, 3, 2, 1, 6, 4, 8, 10, 12, 15, 4, 3 is
 a. 15 b. 14 c. 12 d. 10
- In a frequency distribution with classes 0–10, 10–20 etc., the size of the class intervals is 10. The lower limit of fourth class is
 a. 40 b. 50 c. 20 d. 30

Practice Time

- Find the size of the class interval in the following cases.
 a. 50–55 b. 200–220 c. 0–3
 55–60 220–240 3–6
 60–65 240–260 6–9
 65–70 260–280 9–12
- Given below is the raw data regarding the weights of 30 students.
 62, 37, 50, 54, 61, 41, 47, 51, 59, 60, 68, 63, 38, 46, 58, 52, 68, 51, 68, 60, 69, 63, 59, 53, 60, 58, 59, 65, 68, 58

Prepare a frequency table using an appropriate class interval.

- The following table gives the production of cars in thousands in different years. Represent this data in the form of a bar graph.

Year	2021	2022	2023	2024
Production	46	30	40	38

- Following table gives the number of students studying in a school over the past five years. Represent the data by means of a bar graph.

Year	No. of students
2020	650
2021	725
2022	680
2023	705
2024	690

- Given below is a table showing the number of students fined under different categories last year. Show this information in the form of a pie chart.

Category	No. of students
Late coming	120
Not in uniform	45
Bad behaviour	55
Late fees	75
Total	295

Case-Study Based Question

- Given below are the marks obtained by 40 students of a class.
 30, 61, 55, 56, 60, 25, 49, 46, 28, 56, 42, 33, 62, 55, 74, 31, 42, 56, 62, 62, 62, 54, 68, 47, 56, 76, 67, 39, 61, 50, 52, 43, 36, 35, 79, 26, 39, 47, 45, 52
 a. Find the lowest observation. b. Find the highest observation. c. Find the range.
 d. Find a suitable class interval. e. Prepare a frequency table using tally marks.
 f. Write down the cumulative frequency in the table.

Everyday Maths

- The maximum temperatures on five days of a week are given below. Represent this data in the form of a bar graph.

Day	Maximum temperature (°C)
Monday	38
Tuesday	36
Wednesday	30
Thursday	34
Friday	28

10. Shreya gets ₹1800 as gift. She spends the money in different heads as follows. Represent this information in a pie chart.

Expense	Amount (in ₹)
Books	450
Travelling	250
Entertainment	600
Eating	300
Savings	200
Total	1800

Journal

How does a bar graph differ from a histogram? For which kind of data would a bar graph be preferred over a histogram? Explain with the help of suitable examples.



Cross-Curricular Connect



C-10.1

(Social Science) To enforce the rule of law, India has a judicial system that consists of the mechanism of courts that a citizen can approach when a law is violated. As an organ of government, the judiciary plays a crucial role in the functioning of India's democracy. India has a **single integrated judicial system**. The judiciary in India has a pyramidal structure with the Supreme Court (SC) at the top. High Courts are below the SC, and below them are the district and subordinate courts.

The table given below shows the sanctioned strength and the working strength of the number of judges in India as on 31st August 2022.

Draw a double bar graph to represent the given data.

Number of judges in India

Name of the Court	Sanctioned strength	Working strength
Supreme Court	30	34
High Court	1108	728
District and Subordinate Courts	22,644	17,509

Being Indian



India is known for its rich cultural heritage and diversification is her identity. Indian dance is one of the most revered identities of India's culture, these can be broadly classified into 2 categories – classical dance and folk dance. The classical dance form originated from the Natyashastra and there are 9 classical dance forms in India. The following table lists some of the classical dance forms of India along with the state they belong to:

Dance	Bharatanatyam	Manipuri	Kathak	Odissi	Kathakali	Mohiniattam	Kuchipudi	Sattriya
State	Tamil Nadu	Manipur	Uttar Pradesh	Odisha	Karnataka	Kerala	Andhra Pradesh	Assam

A dance academy has courses available for six classical dance forms. The following table shows the number of students enrolled in various classes. Draw a histogram to represent the given data.

Dance Forms	Bharatanatyam	Manipuri	Kathak	Odissi	Kathakali	Mohiniattam
Number of students	20	24	36	10	18	12

Brain Teaser



- How many students collected 3 coins? _____
 How many students collected 2 coins? _____
 How many students collected 4 coins? _____
 How many students collected 7 coins? _____
 How many students collected more than 8 coins? _____
 How many students were surveyed in all? _____

C-7.2

Coins Collected ○ = 1 Student

